Explanation to Booklet Questions:-

Ex:16:- [B.Q:5]:- A sum of money is to be distributed among A, B, C & D in proportion of 5:2:4:3. If C gets 1000 more than D, what is B's share?

Ans:- A, B, C & D are $5x, 2x, 4x, 3x$.

$C = D + 1000 \Rightarrow 4x = 3x + 1000 \Rightarrow x = 1000$

B's share $= 2x = 2 \times 1000 = 2000$

Ex:17:- [B.Q:6]:- A sum of 312 was devided among 100 students such that Boys gets 3.6/- each & girls gets 2.4/- each. then the no of girls are?

Ans:- Total students = Boys + Girls = $B + G_1 = 100$ — (1)

distributed 312 rs to all 'B' & 'G' like for all Boys each 3.6/- & girls each 2.4/-.

Total boys amount = 3.6/- each × Total Boys = $3.6B$

Total girls " " = 2.4/- " × " Girls = $2.4B$

Total amount = 312 = Total Boys Amount + Total girls amount

$\Rightarrow 312 = 3.6B + 2.4G_1$ — (2)

from (1) & (2):- $3.6B + 2.4G_1 = 312$ — (1)

$B + G_1 = 100$ — (2).

he asked for girls, so, we have to cancel Boys in both.

(1) × 3.6 $\Rightarrow$ $3.6B + 3.6G_1 = 360$

(2) × 1 $\quad 3.6B + 2.4G_1 = 312$

$1.2G_1 = 48 \Rightarrow \boxed{G_1 = 40}$

Boys $= 100 - G_1 = 100 - 40 = 60$

Ex:-18:- [B.Q:8]:- Two numbers 20% & 50% more than the 3rd number respectively. The ratio of 2 no.s is ?

Ans: Assume 3rd number as 'x'.

Ratio of 1st & 2nd = $1.2x : 1.5x = \frac{12}{4} : \frac{15}{5}$

$$\boxed{A : B = 4 : 5}$$

Ex:-19:- [B.Q:-9]:- The salaries of A, B & C are in 2:3:5. If increments of 15%, 10% & 20% allowed then new ratio is ?

Ans:- Lets their salaries = 2x, 3x & 5x.

New Salaries Ratio = $1.15 \times 2x : 1.1 \times 3x : 1.2 \times 5x$

$= 2.3x : 3.3x : 6x = 23 : 33 : 60$

Ex:-20:- [B.Q:-10]:- 18 yrs ago, a father was 3 times as old as his son. Now the father is only twice as old as his son. Then the sum of the present ages of the son & the father is ?

Ans- 18 yrs Ago $\Rightarrow F - 18 = 3(S - 18)$ — ①

present $\Rightarrow F = 2S$ — ②

Sub ② in ① $\Rightarrow 2S - 18 = 3S - 54$

$\Rightarrow 3S - 2S = 54 - 18 \Rightarrow \boxed{S = 36}$

$F = 2S = 2 \times 36 = \boxed{72}$

Ex:-21:- [B.Q:-11]:- The avg of 4 consecutive even nos is 27. Find the largest of these nos?

Ans:- Assume $x, x+2, x+4$ & $x+6$ are 4 consecutive even nos

Avg $= \frac{\text{Sum}}{4} = 27 \Rightarrow$ Sum $= 27 \times 4 = 108$.

$x + x+2 + x+4 + \boxed{x+6} = 4x + 12 = 108 \Rightarrow 4x = 96 \Rightarrow x = 24$

Largest $= x+6 = 24+6 = \boxed{30}$

(109)

Ex:22:- [B.Q:-12]:- A batsman makes a score of 80 runs in his 16th innings & increases his average by 3. What is his avg after 16th innings?

Ans:- Lets assume Avg after 15th innings (or)

Average of 15 innings = x, 16th innings score = 80

Avg after 16th innings (or) Avg of 16 innings = $x+3$

$\therefore$ Avg = $(x+3) = \frac{15x+80}{16} \Rightarrow 16x+48 = 15x+80$

$\Rightarrow 16x-15x = 80-48 \Rightarrow x = 32$

Avg of 16 innings = $x+3 = 32+3 = \underline{35}$

Ex:-23:-[B.Q:-13]:- In an exam, the avg marks of 80 pupils in Class V is 35. The avg of marks in section A of the class is 55, while the average of marks in section B is 30. Find the number of students in class V B?

Ans:- A+B = 80 = Total students in V — (1)

Total marks of V class students in A & B = Avg x No. of

$\Rightarrow 35(A+B) = 80\times35 = 55A+30B$

$\Rightarrow 35A+35B = 55A+30B \Rightarrow 20A = 5B \Rightarrow A = \frac{1}{4}B$ — (2)

Sub (2) in (1) $\Rightarrow A+B = 80 \Rightarrow \frac{1}{4}B+B = 80$

$\frac{5B}{4} = 80 \Rightarrow 5B = 320 \Rightarrow B = \underline{64}$

Ex:-24:- [B.Q:-14]:- The Avg weight of 8 persons goes by 2.5kg when a new person comes in place of one of them weighing 65kg. The weight of new person?

Ans:- Before replacement Avg of 8 assume as x.

Sum before replacement = $8x$, Avg after replace = $x+2.5$

$\Rightarrow$ Avg after = $\frac{8x-65+y\,(\text{replaced by weight})}{8} = x+2.5 \Rightarrow 8x-65+y = 8x+20$

new weight (or) replaced by weight

$\Rightarrow \Rightarrow \boxed{y = 85}$